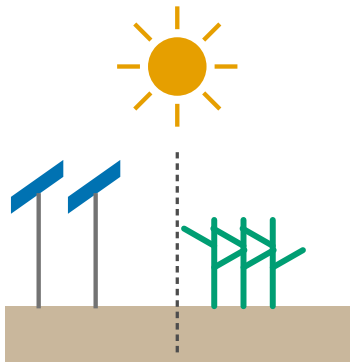
❶ How much solar can share farmland and keep 80% of the harvest?

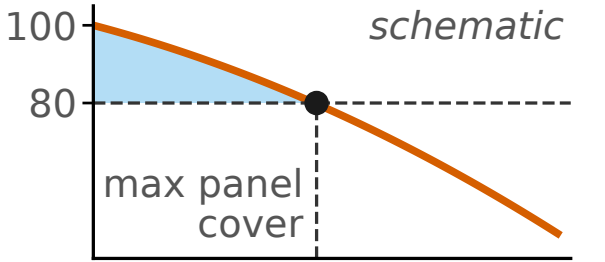


❷ Simulate shade → harvest

shade model → crop model

harvest (% of open field)

schematic

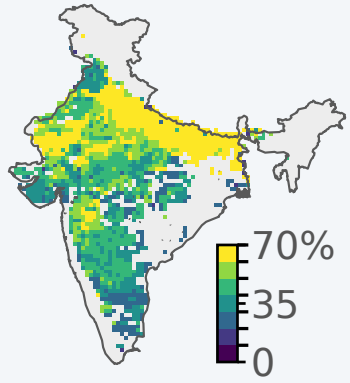


panel cover: sparse → dense

every 0.25° cell · 5 climate models

❸ Answer

max panel cover keeping 80% harvest



45.5 TWp

would generate ≈ 48× India's yearly electricity use

(upper bound)

2050 (SSP5-8.5): [±X]% vs 1995–2014

~61% cereals

~29% legumes